

C++ & the STL

The competitive-coding toolkit · Equigrow short notes

Template

```
#include <bits/stdc++.h>
using namespace std;
int main() {
    ios::sync_with_stdio(false); cin.tie(nullptr);
    int n; cin >> n;
    vector<int> a(n);
    for (auto &x : a) cin >> x;
    sort(a.begin(), a.end());
    cout << a.back() << "\n";
}
```

STL containers

Container	Use	Complexity
vector	dynamic array	push_back $O(1)$
map / set	sorted keys	$O(\log n)$
unordered_map	fast lookup	$O(1)$ average
priority_queue	max-heap (min with greater<>)	$O(\log n)$
deque / stack / queue	both ends / LIFO / FIFO	$O(1)$

Handy

- Use **long long** when values can exceed about 2×10^9 .
- Algorithms: sort, lower_bound, upper_bound, reverse, max_element, accumulate, next_permutation.